



ME 300: Engineering Thermodynamics

Isentropic Relations

Isentropic processes $\hookrightarrow \dot{S}=0$

$$\Delta S = c_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)$$

$\rightarrow$ ideal gas

$\rightarrow c_p = \text{const.}$

$\Rightarrow \Delta S = 0$

$$\left. \begin{array}{l} \rightarrow \text{ideal gas} \\ \rightarrow c_p = \text{const.} \\ \Rightarrow \Delta S = 0 \end{array} \right\} \Rightarrow c_p \ln\left(\frac{T_2}{T_1}\right) = R \ln\left(\frac{P_2}{P_1}\right)$$

$$\frac{c_p}{R} \ln\left(\frac{T_2}{T_1}\right) = \ln\left(\frac{P_2}{P_1}\right)$$

$$\frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^{c_p/R}$$

$$T_1^\gamma P_1^{1-\gamma} = T_2^\gamma P_2^{1-\gamma}$$

ratio of specific heats:

$$\gamma = \frac{c_p}{c_v} > 1$$

$$R = c_p - c_v$$

Isentropic relations

only use if: ideal gas
 $C_p, \gamma = \text{const.}$
 $\Delta S = 0$

$$T_1^\gamma P_1^{1-\gamma} = T_2^\gamma P_2^{1-\gamma}$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\underline{\underline{P_1 V_1^\gamma = P_2 V_2^\gamma}}$$

P-V diagram

$$P V^\gamma = \text{const}$$

$$P V = R T$$

→ isothermal

$$P V = \text{const}$$

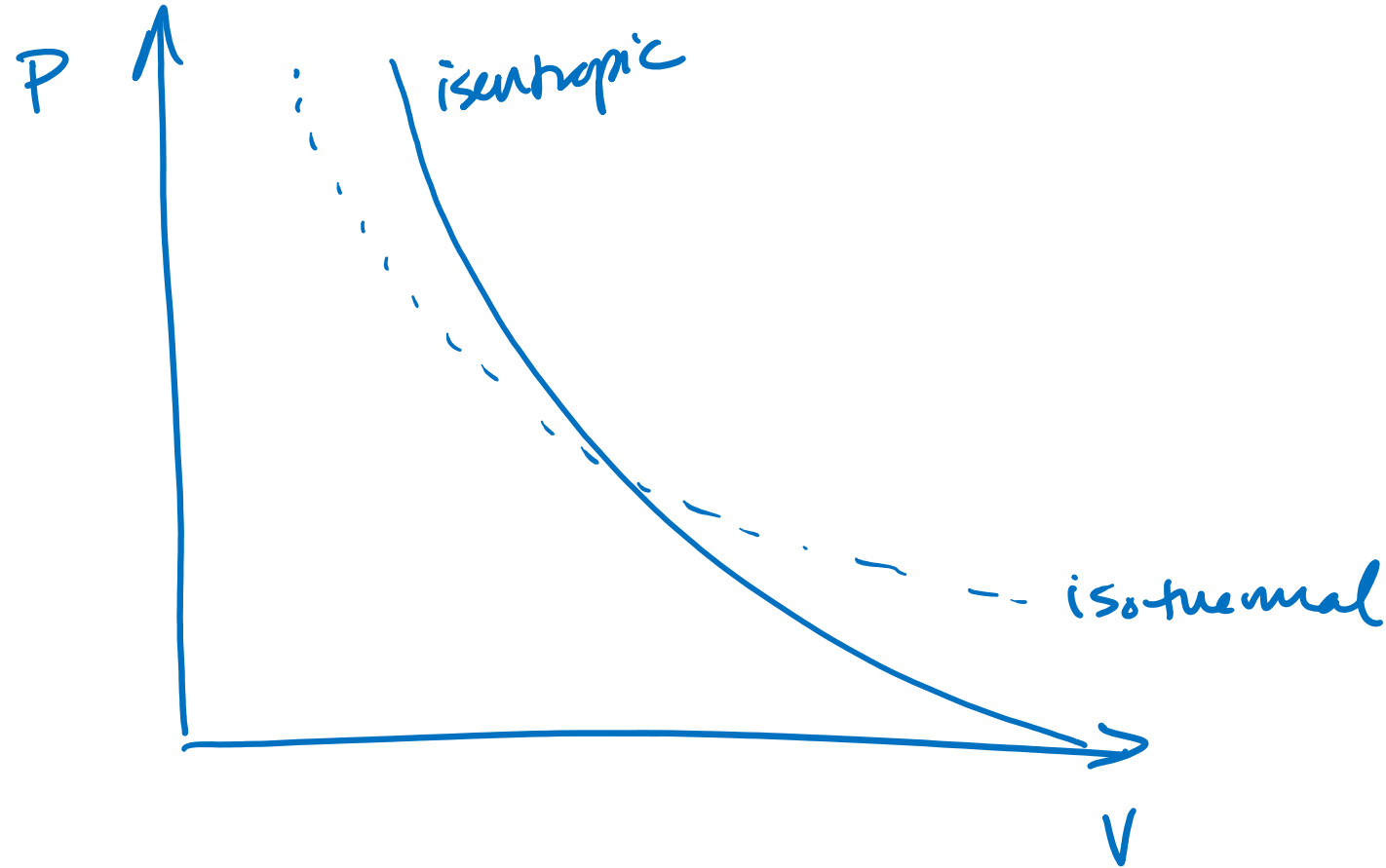
$$P \sim 1/V$$

→ isentropic

$$P V^\gamma = \text{const}$$

$$P \sim 1/V^\gamma$$

$$\gamma > 1 \quad \text{air} \Rightarrow \gamma = 1.4$$



Carnot cycle - reversible

1-2: isothermal heat in $T = \text{const}$, $\frac{P}{R} = \text{const}$

2-3: isentropic expansion $P V^\gamma = \text{const}$

3-4: isothermal heat rejection

4-1: isentropic compression